Colon & Colorectal Cancer — Biomarker Testing Intelligence

Alert Window: Last 30 Days · Generated: 2026-07-09 07:30:03

Situation Summary

Stage IV: 13 of 34
Biomarker Count: 18 entries
Top Targets: MSI/MMR (6) · ctDNA (6) · BRAF (3)
Breakthrough: 12 entries scored ≥ 4

Research & Guidelines Timeline

Biomarker testing entries ranked by significance score. Higher scores indicate greater clinical relevance. Always discuss with your oncologist.

HIGH Stage IV MSI/MMR VERIFIED LIQUID BIOPSY PUBLISHED JUN 25, 2026 10

EMPIRE (NSABP FC-13): a biomarker-driven phase II platform trial evaluating cemiplimab-based immunotherapy in microsatellite-stable colorectal cancer with ctDNA-defined minimal residual disease

In the EMPIRE (NSABP FC-13) phase II trial, cemiplimab-based immunotherapy was evaluated in microsatellite-stable colorectal cancer patients with ctDNA-defined minimal residual disease. The study aims to identify high-risk patients post-curative therapy who may benefit from additional treatment, addressing the limited efficacy of immune checkpoint inhibitors in this population.

pubmed.ncbi.nlm.nih.gov

HIGH Stage IV ctDNA VERIFIED LIQUID BIOPSY PUBLISHED JUN 29, 2026 8

Early circulating tumor DNA dynamics predict fruquintinib efficacy in refractory metastatic colorectal cancer before imaging

Day-28 circulating tumor DNA (ctDNA) dynamics predict efficacy and survival in patients with refractory metastatic colorectal cancer undergoing fruquintinib treatment, enhancing clinical model performance. The study reports a modest discriminative performance (C-index 0.69), indicating the need for prospective validation before clinical application. These findings suggest potential for ctDNA-guided risk stratification in this patient population.

pubmed.ncbi.nlm.nih.gov

HIGH Stage IV ctDNA VERIFIED LIQUID BIOPSY PUBLISHED JUN 29, 2026 6

Breaking news from ESMO 2025: liquid biopsy novelties in gastrointestinal cancer

Liquid biopsy (LB) is advancing in colorectal cancer, particularly with circulating tumor DNA (ctDNA) integration into adjuvant and metastatic treatment decision-making. While LB shows promise in characterizing tumor biology across gastrointestinal malignancies, its clinical applications in other GI tumors remain largely exploratory.

pubmed.ncbi.nlm.nih.gov

HIGH Early Detection ctDNA VERIFIED LIQUID BIOPSY PUBLISHED JUN 12, 2026 6

Diagnostic Accuracy of Blood-Based cfDNA/ctDNA Tests for Colorectal Cancer-A Systematic Review and Meta-Analysis

Blood-based assays for colorectal cancer screening, specifically circulating tumor DNA (ctDNA) and cell-free DNA (cfDNA) tests, show variable diagnostic performance across different platforms. A systematic review and meta-analysis were conducted to assess their accuracy compared to stool-based tests. The findings highlight the need for standardized methodologies to improve the reliability of these non-invasive screening modalities in clinical practice.

pubmed.ncbi.nlm.nih.gov

HIGH Stage III ctDNA VERIFIED LIQUID BIOPSY PUBLISHED JUL 02, 2026 5

When Blood Speaks Before the Scan: The Role of ctDNA in Real-time CRC Care

Serial ctDNA monitoring using Signatera in stage III colorectal cancer facilitated earlier detection of recurrence and informed timely interventions. These results support the clinical integration of ctDNA testing, contingent on further validation through prospective trials.

pubmed.ncbi.nlm.nih.gov

HIGH Early Detection ctDNA VERIFIED LIQUID BIOPSY PUBLISHED JUN 11, 2026 5

A Neural Network-Enabled, Enzymatic cfDNA Methylation Assay for Colorectal Cancer Early Detection

A non-NGS liquid biopsy assay utilizing methylation profiling of circulating cell-free DNA (cfDNA) was developed for early detection of colorectal cancer (CRC). The assay targets 40 CpG regions identified through bioinformatic analysis. Clinical evaluation indicates potential for improved sensitivity and scalability compared to current noninvasive screening methods, addressing limitations in early CRC detection.

pubmed.ncbi.nlm.nih.gov

HIGH All Stages ctDNA VERIFIED LIQUID BIOPSY PUBLISHED JUN 30, 2026 4

Circulating tumour DNA in patients with colorectal cancer undergoing curative-intent LIver MEtastases Surgery in France: a prospective, multicentre GERCOR G-118 CLIMES PRODIGE 77 cohort study protocol

Circulating tumor DNA (ctDNA) is being evaluated for its prognostic value in patients with colorectal cancer liver metastases undergoing curative-intent surgery, as part of a prospective, multicenter GERCOR G-118 CLIMES PRODIGE 77 cohort study. The study aims to enhance the prediction of patient outcomes and the effectiveness of perioperative chemotherapy in this population.

pubmed.ncbi.nlm.nih.gov

MEDIUM All Stages BROAD SCOPE VERIFIED GENOMIC PROFILING PUBLISHED JUN 29, 2026 4

Relationship between MLH1, MSH2, MSH6, and PMS2 protein expression status and clinicopathological characteristics in colorectal cancer tissues

In a cohort of colorectal cancer patients, 22.5% exhibited deficient mismatch repair (dMMR), associated with right-sided tumors, poor differentiation, mucinous histology, larger size, and reduced lymph node metastasis. The study underscores the importance of routine immunohistochemistry (IHC) for MMR status in CRC, while also recommending confirmatory molecular testing methods such as MSI analysis and BRAF V600E mutation assessment.

pubmed.ncbi.nlm.nih.gov

HIGH Stage IV RAS VERIFIED LIQUID BIOPSY PUBLISHED JUN 24, 2026 4

The temporal evaluation of RAS mutation by liquid biopsy at progression after bevacizumab-based treatment in patients with metastatic colorectal cancer

Dynamic changes in plasma RAS mutation status during disease progression in metastatic colorectal cancer (mCRC) were observed, indicating tumor clonal evolution. Notably, RAS mutation clearance correlated with a survival advantage,

suggesting prognostic relevance. Reassessing RAS status via liquid biopsy at progression may yield clinically relevant insights, though findings are exploratory and warrant further investigation.
pubmed.ncbi.nlm.nih.gov

HIGH | Stage III | BROAD SCOPE | VERIFIED | LIQUID BIOPSY | PUBLISHED JUN 12, 2026 | 4

An exploratory multimodal pipeline for recurrence prediction in CRC with XELOX therapy

A multimodal machine learning pipeline was assessed for predicting recurrence in 86 colorectal cancer patients treated with XELOX. The study integrated clinicopathological and liquid biopsy data, identifying 17 recurrences. This exploratory research suggests that combining these data types may enhance risk stratification in CRC recurrence post-surgery and adjuvant therapy.
pubmed.ncbi.nlm.nih.gov

MEDIUM | Stage IV | BROAD SCOPE | VERIFIED | RESEARCH | PUBLISHED JUL 06, 2026 | 2

Genomic Landscape of GNAQ and GNA11 Mutations in Metastatic Solid Tumors: A Real-World Data Analysis

GNAQ and GNA11 mutations are present in various solid tumors, including colorectal cancer, as shown in this real-world data analysis. The study highlights a strong association between nonhotspot mutations and high tumor mutational burden (TMB) or microsatellite instability (MSI), indicating a distinct immunogenic subgroup. This finding emphasizes the importance of distinguishing between canonical hotspot drivers and bystander mutations for optimal selection of targeted therapies versus immune checkpoint inhibitors.
pubmed.ncbi.nlm.nih.gov

MEDIUM | Stage III | MSI/MMR | VERIFIED | RESEARCH | PUBLISHED JUL 06, 2026 | 2

Mismatch repair deficiency or microsatellite instability in locally advanced rectal cancer patients treated with total neoadjuvant therapy

In locally advanced rectal cancer patients undergoing total neoadjuvant therapy, the presence of mismatch repair deficiency (dMMR) or high microsatellite instability (MSI-H) did not demonstrate prognostic significance.
pubmed.ncbi.nlm.nih.gov

MEDIUM | All Stages | BROAD SCOPE | VERIFIED | GENOMIC PROFILING | PUBLISHED JUN 19, 2026 | 2

A systematic scoping review of the molecular pathogenesis of digestive system cancers (2013-2025)

Digestive system cancers (DSCs) are highly prevalent and lethal, comprising over 25% of cancer diagnoses. A systematic scoping review synthesized findings from 57 studies (2013-2025) utilizing next-generation sequencing (NGS) to identify genetic, epigenetic, and transcriptomic alterations across DSCs, including colorectal cancer. The review highlights the need for an integrated overview of molecular pathogenesis to enhance understanding and treatment strategies for these malignancies.
pubmed.ncbi.nlm.nih.gov

MEDIUM | All Stages | MSI/MMR | VERIFIED | RESEARCH | PUBLISHED JUN 15, 2026 | 2

Molecular characteristics of Chinese colorectal cancer patients with microsatellite instability

Chinese colorectal cancer patients with microsatellite instability-high (MSI-H) and microsatellite stable (MSS) tumors exhibit distinct molecular characteristics, including differences in tumor mutational burden (TMB), homologous recombination deficiency (HRD), Epstein-Barr virus (EBV) status, and driver gene alterations. This highlights the molecular heterogeneity of MSI-H colorectal cancer and suggests the necessity for future research incorporating germline and epigenetic data to enhance subtype classification.
pubmed.ncbi.nlm.nih.gov

MEDIUM | Early Detection | BROAD SCOPE | VERIFIED | EARLY DETECTION | PUBLISHED JUN 12, 2026 | 2

Phenotypic heterogeneity in carriers of a pathogenic MSH2 variant: implications for the diagnosis of Lynch syndrome

Carriers of a pathogenic MSH2 variant exhibit phenotypic heterogeneity, impacting Lynch syndrome diagnosis. The study highlights the importance of microsatellite instability (MSI) and immunohistochemistry (IHC) in screening algorithms for Lynch syndrome-associated tumors, which typically show MSI and loss of MMR protein expression. Understanding this heterogeneity may refine diagnostic approaches for patients with suspected Lynch syndrome.

pubmed.ncbi.nlm.nih.gov

MEDIUM All Stages BROAD SCOPE VERIFIED RESEARCH PUBLISHED JUN 30, 2026 1

Superficial serrated adenoma: a clinicopathological analysis of ten cases

Superficial serrated adenomas (SuSAs) exhibit distinct clinicopathological features, as analyzed in ten cases from a single institution. Histological morphology and immunophenotype were assessed, focusing on markers such as CK7, CK20, Ki-67, β-catenin, C-MYC, and mismatch repair proteins. The findings contribute to the understanding of SuSA's diagnostic characteristics, potentially aiding in the differentiation of these lesions from other colorectal neoplasms.

pubmed.ncbi.nlm.nih.gov

HIGH Stage IV BROAD SCOPE VERIFIED DIAGNOSTIC APPROVAL PUBLISHED JUN 22, 2026 1

Quantitation of Tucatinib, a Novel Tyrosine Kinase Inhibitor in Dried Blood Spot (DBS) With LC-ESI-MS/MS: Application to a Pharmacokinetic Study in Mice

Tucatinib, a selective HER2 kinase inhibitor, was quantified in dried blood spots using LC-ESI-MS/MS in a pharmacokinetic study in mice. The study supports the use of Tucatinib as a targeted therapy for HER2-positive metastatic colorectal cancer, demonstrating its potential to improve survival and disease control, particularly in challenging cases with brain metastases.

pubmed.ncbi.nlm.nih.gov

MEDIUM Stage III BROAD SCOPE VERIFIED RESEARCH PUBLISHED JUN 16, 2026 1

Severe renal toxicity following adjuvant envafolimab in a patient with ultra-hypermutated (POLE) stage II colorectal cancer: a case report

Ultra-hypermutated colorectal cancer (CRC) with POLE mutation may be associated with increased risk of immune-related renal toxicity following adjuvant envafolimab. This case highlights the need for careful patient selection, thorough risk-benefit assessment, and vigilant monitoring of organ function in early-stage POLE-mutated CRC patients receiving off-label immunotherapy, as the potential for unnecessary toxicity exists despite favorable prognosis.

pubmed.ncbi.nlm.nih.gov

MEDIUM All Stages BRAF VERIFIED RESEARCH PUBLISHED JUN 12, 2026 1

Molecular Landscape and Advanced Diagnostic Technologies for BRAF Mutations in Cancer: From Quantitative PCR and ddPCR to CRISPR-Based Platforms

BRAF mutations, particularly BRAF-V600E, are significant in colorectal cancer and other malignancies, driving MAPK pathway activation and impacting treatment response. The study discusses advanced diagnostic technologies, including quantitative PCR, ddPCR, and CRISPR-based platforms, for accurate detection of BRAF alterations, underscoring their importance in molecular classification and prognostic evaluation.

pubmed.ncbi.nlm.nih.gov

Appendix A — Patient-Friendly Summary

Plain-language overview for patients, caregivers, and family members

What's happening

Biomarker testing — also called molecular profiling or genomic testing — analyzes your tumor's DNA to identify specific mutations. The results determine which FDA-approved treatments may be available to you. The entries below represent the highest-scoring study per biomarker target drawn from the full research archive — not limited to the current report window. Some entries may predate the timeline above; they are included because they remain the most clinically significant reference available for that biomarker.

MSI/MMR

EMPIRE (NSABP FC-13): a biomarker-driven phase II platform trial evaluating cemiplimab-based immunotherapy in microsatellite-stable colorectal cancer with ctDNA-defined minimal residual disease

In the EMPIRE (NSABP FC-13) phase II trial, cemiplimab-based immunotherapy was evaluated in microsatellite-stable colorectal cancer patients with ctDNA-defined minimal residual disease. The study aims to identify high-risk patients post-curative therapy who may benefit from additional treatment, addressing the limited efficacy of immune checkpoint inhibitors in this population.

KRAS

Circulating tumor DNA (ctDNA) clearance as an early indicator of sotorasib + panitumumab efficacy and prognosis in KRAS G12C–mutated colorectal cancer (mCRC): Results from phase 3 CodeBreaK 300.

In the phase 3 CodeBreaK 300 trial, ctDNA clearance was assessed as an early indicator of treatment efficacy and prognosis in KRAS G12C–mutated mCRC patients receiving sotorasib and panitumumab. The findings suggest that ctDNA clearance may serve as a valuable biomarker for monitoring therapeutic response and predicting outcomes in this patient population.

ctDNA

Circulating tumor DNA-based assessment of disease progression after liver resection or transplantation for metastatic colorectal cancer or hepatocellular carcinoma-study protocol for a randomized, controlled trial

Circulating tumor DNA (ctDNA) monitoring is being evaluated for its clinical utility in detecting minimal residual disease (MRD) after liver resection or transplantation for metastatic colorectal cancer (CRLM) and hepatocellular carcinoma (HCC). This randomized controlled trial aims to determine if ctDNA can provide earlier detection of disease progression compared to conventional imaging and biomarkers, which have limited sensitivity.

TP53

Clinical impact of TP53 mutation status on survival outcomes in metastatic colorectal cancer

TP53 mutations serve as an independent prognostic marker for prolonged progression-free survival in metastatic colorectal cancer. While overall survival differences were not statistically significant, the findings suggest the need for further validation in prospective studies to assess TP53's role in therapeutic stratification.

BRAF

Detection of KRAS, NRAS and BRAF Mutations in Liquid Biopsy from Patients with Colorectal Cancer

ddPCR analysis of circulating tumor DNA (ctDNA) in colorectal cancer patients reveals that it can provide complementary information for molecular diagnosis. The study highlights the utility of liquid biopsy in detecting KRAS, NRAS, and BRAF mutations, enhancing diagnostic accuracy in this patient population.

NGS

Rapid On-Site Next-Generation Sequencing: An Alternative to Single-Gene and Send-Out Testing in Non-Small Cell Lung Cancer and Colorectal Cancer in a Community Pathology Laboratory Setting

Next-generation sequencing (NGS) using the ThermoFisher Oncomine Precision Assay (OPA) was assessed as an alternative to single-gene panels and send-out NGS for identifying actionable mutations in colorectal cancer. The study focused on turnaround time, quantity not sufficient rates, and detection of NCCN-recommended alterations, indicating that OPA may enhance efficiency in mutation identification for targeted therapies in a community pathology laboratory setting.

RAS

The temporal evaluation of RAS mutation by liquid biopsy at progression after bevacizumab-based treatment in patients with metastatic colorectal cancer

Dynamic changes in plasma RAS mutation status during disease progression in metastatic colorectal cancer (mCRC) were observed, indicating tumor clonal evolution. Notably, RAS mutation clearance correlated with a survival advantage, suggesting prognostic relevance. Reassessing RAS status via liquid biopsy at progression may yield clinically relevant insights, though findings are exploratory and warrant further investigation.

HER2

Comprehensive genomic landscape of ERBB2 in Chinese GI tumors: mutation-centered landscapes and precision treatment opportunities

ERBB2 alterations in colorectal cancer (CRC) and gastric cancer (GC) predominantly occur in kinase domain hotspots, influenced by tumor-specific genomic contexts. Oncogenic ERBB2 mutations correlate with higher tumor mutational burden (TMB) and microsatellite instability-high (MSI-H) status. These findings support the clinical evaluation of mutation-selective HER2 inhibitors and their potential combination with immune checkpoint blockade for targeted therapy in CRC and GC.

Genomic Profiling

Integrated Molecular Profiling of Colorectal Cancer by Tumor Location: Evidence from a Real-World Cohort with Primary and Metastatic Samples

Colorectal cancer (CRC) tumor location correlates with distinct molecular profiles. Right-sided tumors exhibit dMMR, MSI-H, and elevated TMB, while left-sided tumors show increased ERBB2 alterations and class 5 APC mutations. These findings underscore the necessity of incorporating tumor location into personalized molecular diagnostics and treatment strategies for CRC patients.

EGFR

Circulating Butyrate Attenuates Cetuximab Efficacy in Colorectal Cancer Through EGFR and AMPK-Wip1 Signaling

Butyrate may serve as a predictive biomarker for treatment response in KRAS wild-type colorectal cancer, as it reduces cetuximab efficacy via EGFR and AMPK-Wip1 signaling pathways.

miRNA

Diagnostic Performance of Circulating miRNA-92a and Related Circulating miRNA Panels for Colorectal Cancer: An Updated Systematic Review and Meta-Analysis

Circulating miRNA-92a shows superior diagnostic performance as a non-invasive biomarker for early colorectal cancer detection. The systematic review and meta-analysis suggest its potential utility, though further large-scale prospective studies are needed to standardize testing protocols and confirm clinical applicability.

TMB

Plasma-TMB (pTMB) and circulating tumor fraction (ctF) dynamics as biomarkers of benefit from the addition of atezolizumab to first-line FOLFOXIRI/bevacizumab in metastatic colorectal cancer (mCRC): A translational analysis of the AtezoTRIBE study.

In the AtezoTRIBE study, plasma tumor mutational burden (pTMB) and circulating tumor fraction (ctF) were analyzed as biomarkers for the efficacy of adding atezolizumab to first-line FOLFOXIRI/bevacizumab in metastatic colorectal cancer. The study suggests that pTMB and ctF dynamics may predict treatment benefit, providing insights for personalized therapy strategies in mCRC.

Stool DNA

Enhancing non-invasive colorectal cancer screening with stool DNA methylation markers and light gradient-boosting machine learning

Stool DNA methylation markers CNRIP1, SFRP2, and VIM demonstrated high sensitivity for non-invasive colorectal cancer screening. The application of the LightGBM machine learning algorithm improved diagnostic accuracy, suggesting a viable method for early CRC detection.

Methylation

Methylation biomarkers for early detection of colorectal cancer: From molecular discovery to clinical translation and application

Key methylation biomarkers (SEPT9, BMP3, NDRG4) have been identified for early detection of colorectal cancer (CRC) through genome-wide association studies and candidate gene approaches. These biomarkers have been validated using high-throughput technologies, facilitating their transition from research to clinical application. Early screening utilizing these methylation markers may improve patient outcomes in CRC management.

IHC

Integration of deep learning and radiomic features from multiplex immunohistochemistry images for reproducible Multi-Outcome prediction in a Multi-Center study of colorectal cancer

Fused multiplex immunohistochemistry (mIHC)-derived radiomic and deep learning features provide accurate and interpretable predictions for multiple colorectal cancer outcomes. The study supports the integration of these features into precision oncology workflows, enhancing the potential for reproducible multi-outcome predictions across different centers.

What this means for you

If you have been diagnosed with colorectal cancer, ask your oncologist: *"Has my tumor been tested for all the biomarkers that have an FDA-approved treatment?"* This is especially important if you have been diagnosed with metastatic (Stage IV) colorectal cancer, where biomarker results directly determine which targeted therapies and immunotherapies are available to you. Based on the research in this report, the most actively studied biomarkers right now are **MSI/MMR, KRAS, ctDNA, TP53, and BRAF**. Testing should happen at diagnosis — before treatment begins — because results take 1–2 weeks and can shape your first treatment decision.

Generated from structured biomarker research data · 2026-07-09 07:30:03
This is not medical advice. Always speak with your oncologist before making any treatment decisions.

Appendix B — Colon & Colorectal Cancer: Biomarker Testing Reference

What Is Biomarker Testing?

Biomarker testing — also called molecular profiling or genomic testing — analyzes your tumor's DNA to identify specific mutations. A single next-generation sequencing (NGS) panel can check many gene mutations at once. Results typically take 1–2 weeks and should be obtained at diagnosis, before treatment begins.

Key Biomarkers — Test All mCRC at Diagnosis

RAS (KRAS/NRAS) · BRAF V600E · MSI/MMR status · HER2 · NTRK fusion · TMB

Testing Methods

NGS (next-generation sequencing) — checks many mutations at once from tumor tissue

Liquid biopsy / ctDNA — blood test that detects tumor DNA; useful when tissue is unavailable

IHC (immunohistochemistry) — used to test MMR status and HER2

PCR — used for specific mutation detection (e.g. MSI, KRAS)

Questions to Ask Your Oncologist

"Has my tumor been tested for all the biomarkers that have an FDA-approved treatment?"

"Was my sample taken from the primary tumor or a metastatic site — and does it need to be retested?"

"How long will results take, and will they affect my first treatment decision?"

"Is a liquid biopsy appropriate for my situation?"

Abbreviations

BRAF V600E — a specific gene mutation found in ~8–10% of colorectal cancers

ctDNA — circulating tumor DNA; tumor DNA fragments found in the bloodstream

dMMR — mismatch repair deficient; indicates the tumor may respond to immunotherapy

HER2 — a protein that drives growth in ~2–3% of colorectal cancers

IHC — immunohistochemistry; a lab staining method used to detect proteins in tissue

KRAS G12C — a specific gene mutation found in ~3–4% of colorectal cancers

mCRC — metastatic colorectal cancer (Stage IV)

MSI-H — microsatellite instability-high; indicates the tumor may respond to immunotherapy

NGS — next-generation sequencing; a method that checks many gene mutations at once

NTRK — a rare gene fusion found in <1% of colorectal cancers

RAS — a gene family (KRAS/NRAS) whose mutation status affects treatment selection

TMB — tumor mutational burden; a measure of how many mutations are in a tumor

Key Resources

ClinicalTrials.gov · NCCN Guidelines (nccn.org) · Colorectal Cancer Alliance (ccalliance.org) · Fight Colorectal Cancer (fightcrc.org) · ACS (cancer.org)

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Limitations

This report does not represent a complete picture of all published research. Database coverage, feed availability, and relevance filtering affect what appears. Research findings may not yet reflect current clinical guidelines. Always verify information at the original source before making clinical decisions.

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